



**AG00686 BMC 6Cyln.**

**ACC-Drive CRK-ø172; AC-ø137**

**Tensioner T38624; CW; 22.2Nm**

**8PK1520HD Fleetranner**

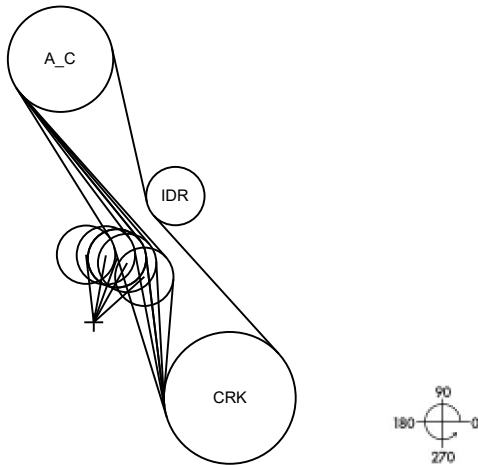
**07Sep2023**

**Belt Data, DB Ver. = 2.18.0.0, Belt Rev. = 7/14/2016**

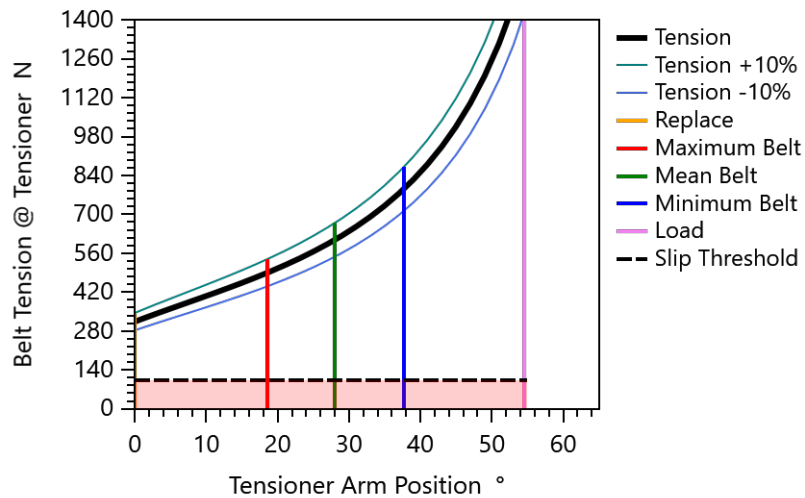
|                                     |               |
|-------------------------------------|---------------|
| Belt Code                           | MT610         |
| # of Ribs / Cord Material           | 8 / polyester |
| Effective Belt Length mm (ISO 9981) | 1520          |

### Tensioner Data

|                  |               |
|------------------|---------------|
| Type             | Automatic     |
| Design Tension N | 609 (137 lbf) |



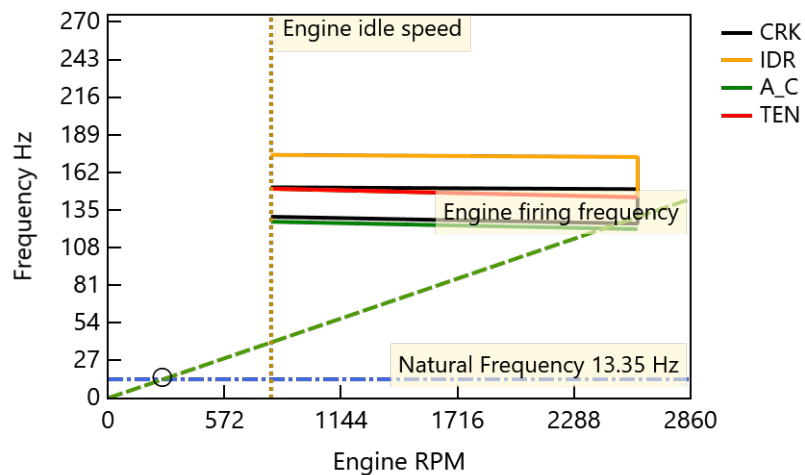
### Belt Tension Control



### Belt Life B10

| Duty Cycle<br>DCn-th % | Av. Vehicle<br>Speed km/hr | PK-2_2p-MT3<br>hours |
|------------------------|----------------------------|----------------------|
| 95                     | 1                          | 13310                |

### Natural Frequency Map



### Peak Tension & Hubload

|     | Engine<br>RPM | Accel.<br>RPM/s | Tension<br>N | Hubload<br>N | Direction<br>/Wrap ° |
|-----|---------------|-----------------|--------------|--------------|----------------------|
| CRK | 1157          | 1000            | 947          | 1486.8       | 119 / 215            |
| IDR | 1157          | 1000            | 944          | 480.0        | 28 / 29              |
| A_C | 1157          | 1000            | 656          | 1559.4       | 294 / 206            |
| TEN | 800           | 1000            | 609          | 353.6        | 210 / 32             |

Note: Peak Loads are defined as the highest loads (tension, shear stress and hubload) calculated for all peak load & accelerations conditions. (Not from system vibration calculation)

### Permissible Grooved Pulley

#### Axial Offset Allowance

For 0.90° Belt Entry Angle

| Adjacent<br>Grooved<br>Pulleys | Flat<br>Pulley<br>Angular<br>Misalignment<br>° | Axial<br>Offset<br>Allowance<br>mm | Grooved<br>Pulley<br>Axial<br>Offset<br>% | Attributed to<br>Flat Pulley<br>Angular<br>Misalignment<br>% |
|--------------------------------|------------------------------------------------|------------------------------------|-------------------------------------------|--------------------------------------------------------------|
| A_C->CRK                       | IDR (0.33)                                     | 3.89                               | 74                                        | 26                                                           |
| CRK->A_C                       | TEN (1.20)                                     | 1.8                                | 10                                        | 90                                                           |



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**07Sep2023**

### Layout Data mm

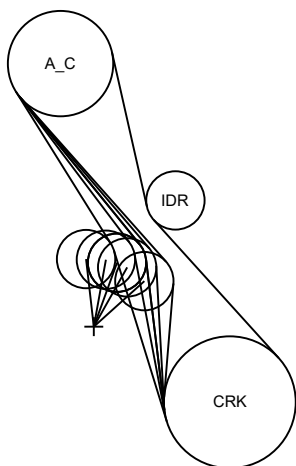
|     | X       | Y      | Flat  | Pitch  | Effective | DOB    |
|-----|---------|--------|-------|--------|-----------|--------|
| CRK | 0.00    | 0.00   |       | 174.40 | 172.00    | 172.99 |
| IDR | -72.00  | 267.00 | 75.00 | 77.20  | 79.60     |        |
| A_C | -224.00 | 448.00 |       | 139.40 | 137.00    | 137.99 |
| TEN | -149.27 | 184.59 | 75.00 | 77.20  | 79.60     |        |

### Belt Data, DB Ver. = 2.18.0.0, Belt Rev. = 7/14/2016

|                                        |               |
|----------------------------------------|---------------|
| Belt Code                              | MT610         |
| # of Ribs / Cord Material              | 8 / polyester |
| Stretch and Wear Allowance % of Length | 0.70          |
| Belt Length Tolerance (+/-) mm         | 6.00          |

### Tensioner Data

|                                   |                  |
|-----------------------------------|------------------|
| Type                              | Automatic        |
| Design Tension N                  | 609 (137 lbf)    |
| Pivot Point {X, Y Coordinates} mm | -180.00 , 100.00 |
| Arm Length mm                     | 90.0             |
| Spring Mean Load Nm               | 22.20            |
| Spring Pre-Load Nm                | 8.86             |
| Spring Rate Nm/deg                | 0.476            |

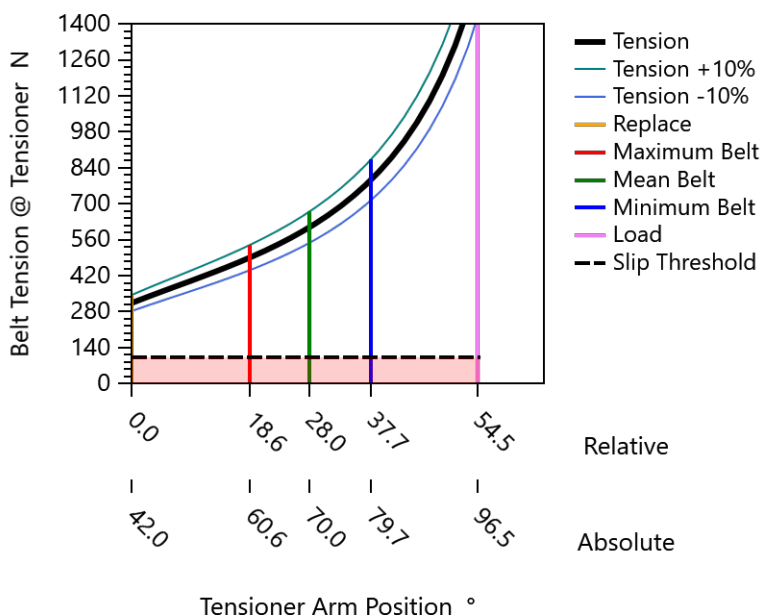


## RESULTS

### Belt Drive System Geometry

|     | Span Length<br>mm | Wrap Angle<br>° | Speed Ratio<br>(Ref. Engine) |
|-----|-------------------|-----------------|------------------------------|
| CRK | 246.3             | 215.2           | 1.000                        |
| IDR | 210.1             | 29.4            | 2.259                        |
| A_C | 251.5             | 206.4           | 1.251                        |
| TEN | 201.3             | 32.2            | 2.259                        |

### Belt Tension Control





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**ACC-Drive CRK-ø172; AC-ø137**  
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**8PK1520HD Fleetrunner**  
**07Sep2023**

### Tensioner Geometry

Belt Takeup / Tensioner Arm Ratio mm/°

0.636

| Position Description         | Belt-Drive Length |         |        |        |        |        |
|------------------------------|-------------------|---------|--------|--------|--------|--------|
|                              | Free Arm          | Replace | Max    | Mean   | Min    | Load   |
| Arm Position °               | 42.0              | 42.0    | 60.6   | 70.0   | 79.7   | 96.5   |
| Idler X Coordinate mm        | -113.1            | -113.1  | -135.9 | -149.3 | -164.0 | -190.2 |
| Idler Y Coordinate mm        | 160.2             | 160.2   | 178.4  | 184.6  | 188.6  | 189.4  |
| Hubload to Arm Angle °       | 23.5              | 23.5    | 38.8   | 47.0   | 55.8   | 71.9   |
| Hubload Direction °          | 198.5             | 198.5   | 201.9  | 203.0  | 203.9  | 204.6  |
| Hubload N                    | 246.8             | 246.8   | 314.6  | 337.4  | 360.2  | 406.8  |
| Tension N                    | 313.8             | 313.8   | 490.7  | 608.8  | 793.4  | 1579.2 |
| Idler Wrap Angle °           | 46.3              | 46.3    | 37.4   | 32.2   | 26.2   | 14.8   |
| Effective Drive Length mm    | 1538.4            | 1538.4  | 1527.7 | 1521.7 | 1515.7 | 1507.3 |
| Required Eff. Belt Length mm | 1536.3            | 1536.3  | 1525.6 | 1519.6 | 1513.6 | 1505.2 |

### Belt Take-up





**AG00686 BMC 6Cyln.**

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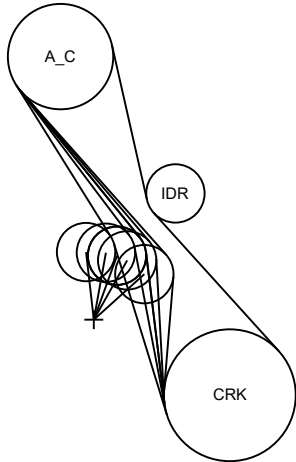
**Tensioner T38624; CW; 22.2Nm**

**8PK1520HD Fleetranner**

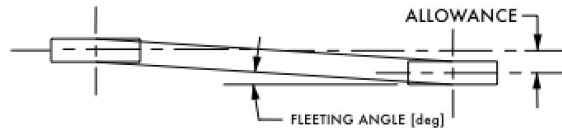
**07Sep2023**

## Alignment Data °

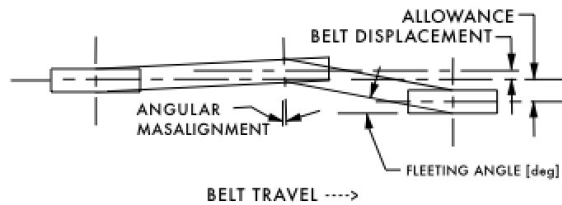
|                          |      |
|--------------------------|------|
| Fleeting Angle           | 0.90 |
| IDR Angular Misalignment | 0.33 |
| TEN Angular Misalignment | 1.20 |



## (GROOVED - GROOVED) PULLEYS



## (GROOVED - FLAT - GROOVED) PULLEYS



## RESULTS

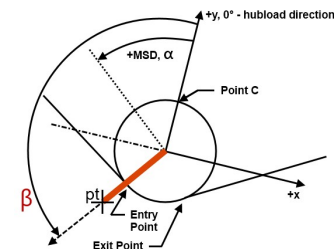
### Permissible Grooved Pulley

#### Axial Offset Allowance

For 0.90° Belt Entry Angle

| Adjacent Grooved Pulleys | Flat Pulley Angular Misalignment ° | Axial Offset Allowance mm | Grooved Pulley Axial Offset % | Attributed to Flat Pulley Angular Misalignment % |
|--------------------------|------------------------------------|---------------------------|-------------------------------|--------------------------------------------------|
| A_C->CRK                 | IDR (0.33)                         | 3.89                      | 74                            | 26                                               |
| CRK->A_C                 | TEN (1.20)                         | 1.8                       | 10                            | 90                                               |

### Hubload to Arm Angle & MSD [°]



Hubload-to-arm angle is the angle between hubload and arm where arm direction is from pulley center to pivot.

β = Hubload-to-arm angle  
pt = Tensioner pivot point

### Backside Pulley Alignment Sensitivity

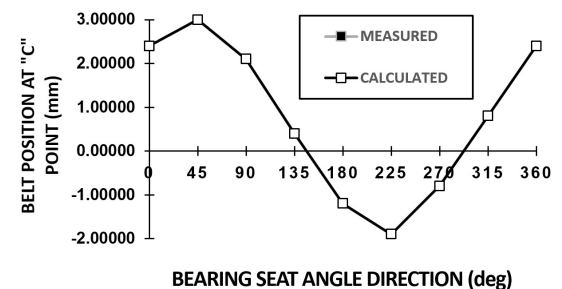
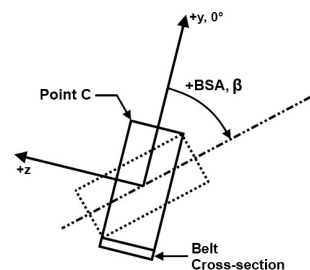
| Backside Pulley | Angular Misalignment | Most Sensitive Direction | Resulting Fleeting Angle | Suggestions |
|-----------------|----------------------|--------------------------|--------------------------|-------------|
| 2: IDR          | 0.330                | 69.55                    | 0.23                     | AB C E      |
| 4: TEN          | 1.200                | 69.64                    | 0.81                     | AB C E      |

Suggestion List for Backside Pulley Alignment Sensitivity

- A] Increase EXIT span length
- B] Decrease ENTRANCE span length
- C] Increase both ENTRANCE / EXIT span lengths
- D] Increase wrap angle
- E] Decrease wrap angle

Note: Most Sensitive Direction for Grooved pulleys is parallel to the belt entry span.

### Direction of Bearing Seat Angle [°]





User: Abdurrahman Kol

Design: AG00686\_BMC\_6Cyln\_ACC-Drive\_CRKø172-ACø137\_8PK1520HD\_T38624-22.2Nm\_CW\_07Sep2023

**AG00686 BMC 6Cyln.**

**ACC-Drive CRK-ø172; AC-ø137**

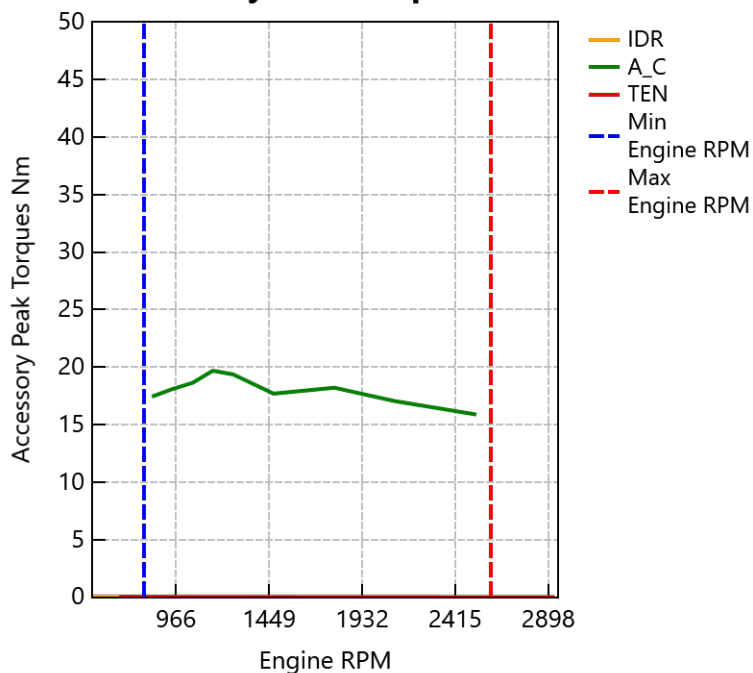
**Tensioner T38624; CW; 22.2Nm**

**8PK1520HD Fleetranner**

**07Sep2023**

|                           |                 |
|---------------------------|-----------------|
| Design Tension N          | 609 (137 [lbf]) |
| Engine Acceleration RPM/s | 1000            |
| Engine Deceleration RPM/s | 1000            |

## Accessory Peak Torques Nm



## Accessory Data

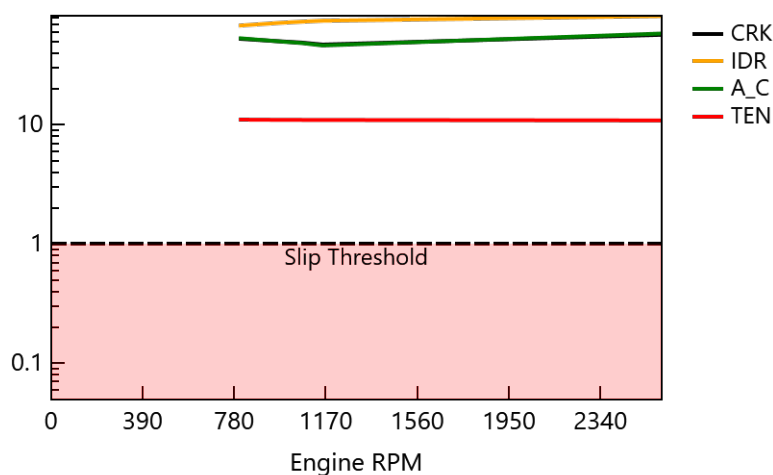
|     | File Name - if used | Moment of Inertia kg m <sup>2</sup> |
|-----|---------------------|-------------------------------------|
| IDR | IDR.IDR             | 0.0004                              |
| A_C | AC.A_C              | 0.0030                              |
| TEN | TEN.TEN             | 0.0076                              |

## RESULTS

### Belt Slip Sensitivity

| Pulley | Most Critical Load Condition |                    |     |  |  | Required Increase Tension or Wrap Angle |   |
|--------|------------------------------|--------------------|-----|--|--|-----------------------------------------|---|
|        | Acceler. RPM/s               | Load Combination % | A_C |  |  | N                                       | ° |
| CRK    | 1000                         | 100                |     |  |  |                                         |   |
| IDR    | 1000                         | 10                 |     |  |  |                                         |   |
| A_C    | -1000                        | 100                |     |  |  |                                         |   |
| TEN    | -1000                        | 100                |     |  |  |                                         |   |

### Belt Slip Safety Factor





User: Abdurrahman Kol

Design: AG00686\_BMC\_6Cyln\_ACC-Drive\_CRKø172-ACø137\_8PK1520HD\_T38624-22.2Nm\_CW\_07Sep2023

**AG00686 BMC 6Cyln.**

**ACC-Drive CRK-ø172; AC-ø137**

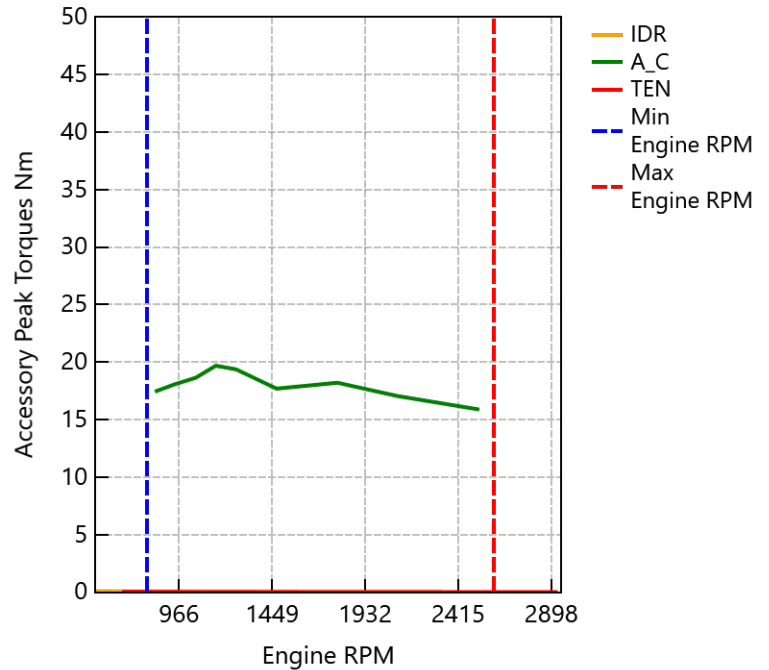
**Tensioner T38624; CW; 22.2Nm**

**8PK1520HD Fleetranner**

**07Sep2023**

|                           |                 |
|---------------------------|-----------------|
| Design Tension N          | 609 (137 [lbf]) |
| Engine Acceleration RPM/s | 1000            |
| Engine Deceleration RPM/s | 1000            |

## Accessory Peak Torques Nm



## RESULTS

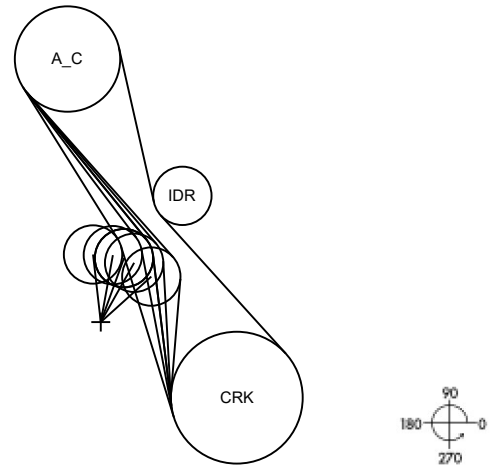
### Peak Tension & Hubload

|     | Engine RPM | Accel. RPM/s | Tension N | Hubload N | Direction /Wrap ° | Shear Stress N/mm-rib @ RPM |
|-----|------------|--------------|-----------|-----------|-------------------|-----------------------------|
| CRK | 1157       | 1000         | 947       | 1486.8    | 119 / 215         | 0.133 @ 1157                |
| IDR | 1157       | 1000         | 944       | 480.0     | 28 / 29           | N/A                         |
| A_C | 1157       | 1000         | 656       | 1559.4    | 294 / 206         | 0.149 @ 1157                |
| TEN | 800        | 1000         | 609       | 353.6     | 210 / 32          | N/A                         |

Note: Peak Loads are defined as the highest loads (tension, shear stress and hubload) calculated for all peak load & accelerations conditions. (Not from system vibration calculation)



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**8PK1520HD Fleetrunner**  
**07Sep2023**



**Load Conditions kW for DC 95%**

| %     | Eng.<br>RPM | V,<br>km/hr | T,<br>°C | CRK  | IDR  | A_C  | TEN  |
|-------|-------------|-------------|----------|------|------|------|------|
| 27.00 | 800         | 1           | 92       | 3.02 | 0.01 | 3.00 | 0.01 |
| 10.00 | 1000        | 1           | 92       | 4.02 | 0.01 | 4.00 | 0.01 |
| 13.00 | 1250        | 1           | 92       | 5.02 | 0.01 | 5.00 | 0.01 |
| 18.00 | 1500        | 1           | 92       | 5.52 | 0.01 | 5.50 | 0.01 |
| 19.00 | 1750        | 1           | 92       | 6.82 | 0.01 | 6.80 | 0.01 |
| 13.00 | 2000        | 1           | 92       | 8.22 | 0.01 | 8.20 | 0.01 |

**Mean Hubloads N for DC 95%**

| %     | Eng<br>RPM | CRK  | IDR | A_C  | TEN |
|-------|------------|------|-----|------|-----|
| 27.00 | 800        | 1560 | 518 | 1591 | 338 |
| 10.00 | 1000       | 1586 | 532 | 1617 | 338 |
| 13.00 | 1250       | 1585 | 532 | 1617 | 338 |
| 18.00 | 1500       | 1550 | 513 | 1580 | 338 |
| 19.00 | 1750       | 1573 | 525 | 1604 | 338 |
| 13.00 | 2000       | 1595 | 537 | 1627 | 338 |

**Mean Tensions N for DC 95%**

| %     | Eng<br>RPM | CRK  | IDR  | A_C | TEN |
|-------|------------|------|------|-----|-----|
| 27.00 | 800        | 1022 | 1021 | 610 | 609 |
| 10.00 | 1000       | 1049 | 1048 | 610 | 609 |
| 13.00 | 1250       | 1049 | 1048 | 610 | 609 |
| 18.00 | 1500       | 1012 | 1011 | 610 | 609 |
| 19.00 | 1750       | 1036 | 1035 | 609 | 609 |
| 13.00 | 2000       | 1059 | 1058 | 609 | 609 |



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**07Sep2023**

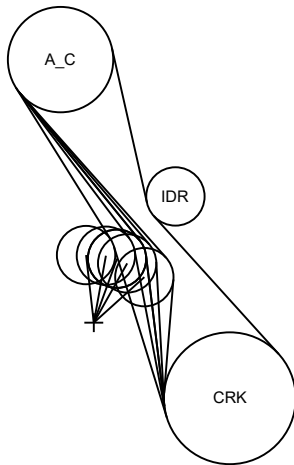
### Load Conditions [kW]

**Belt Data, DB Ver. = 2.18.0.0, Belt Rev. = 7/14/2016**

|                           |               |
|---------------------------|---------------|
| Belt Code                 | MT610         |
| # of Ribs / Cord Material | 8 / polyester |

### Tensioner Data

|                  |               |
|------------------|---------------|
| Type             | Automatic     |
| Design Tension N | 609 (137 lbf) |







AG00686 BMC 6Cyln.

ACC-Drive CRK-ø172; AC-ø137

Tensioner T38624; CW; 22.2Nm

8PK1520HD Fleetrunner

07Sep2023

## RESULTS

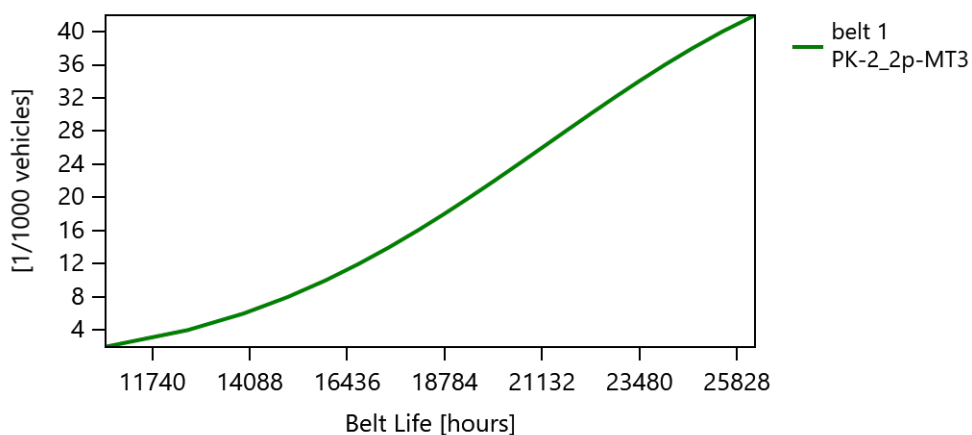
### Pulley Contributions to Belt Rib Fatigue %

|             | CRK | IDR  | A_C | TEN  |
|-------------|-----|------|-----|------|
| PK-2_2p-MT3 | 1.9 | 45.3 | 7.4 | 45.3 |

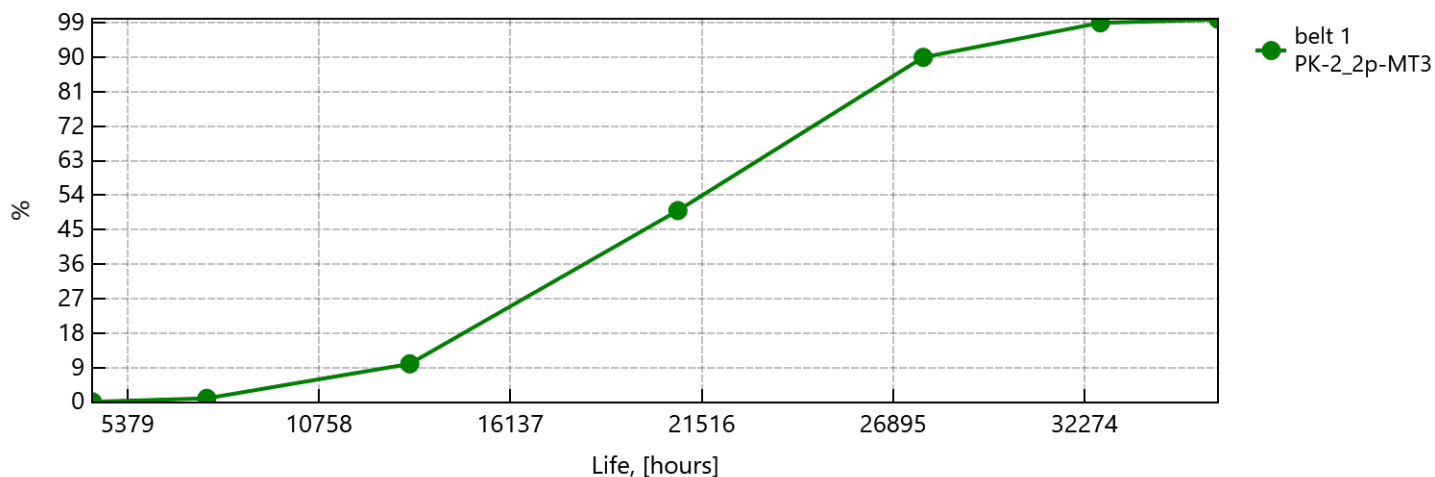
### Belt Life B10

| Duty Cycle<br>DCn-th % | Av. Vehicle<br>Speed km/hr | PK-2_2p-MT3<br>hours |
|------------------------|----------------------------|----------------------|
| 95                     | 1                          | 13310                |

### Belt Incidents per Thousand Vehicles



### Weibull Distribution for DC95%



### Load Conditions kW for DC 95%

| %     | Eng.<br>RPM | V,<br>km/hr | T,<br>°C | CRK  | IDR  | A_C  | TEN  |
|-------|-------------|-------------|----------|------|------|------|------|
| 27.00 | 800         | 1           | 92       | 3.02 | 0.01 | 3.00 | 0.01 |
| 10.00 | 1000        | 1           | 92       | 4.02 | 0.01 | 4.00 | 0.01 |
| 13.00 | 1250        | 1           | 92       | 5.02 | 0.01 | 5.00 | 0.01 |
| 18.00 | 1500        | 1           | 92       | 5.52 | 0.01 | 5.50 | 0.01 |
| 19.00 | 1750        | 1           | 92       | 6.82 | 0.01 | 6.80 | 0.01 |
| 13.00 | 2000        | 1           | 92       | 8.22 | 0.01 | 8.20 | 0.01 |

### Load Contribution to Belt Rib Fatigue for DC 95%

| %     | Vblt<br>m/s | CRK<br>Tt N | CRK<br>Ts N | %<br>PK-2_2p-MT3 |
|-------|-------------|-------------|-------------|------------------|
| 27.00 | 7.3         | 1022        | 609         | 17               |
| 10.00 | 9.1         | 1049        | 609         | 8                |
| 13.00 | 11.4        | 1049        | 609         | 12               |
| 18.00 | 13.7        | 1012        | 609         | 20               |
| 19.00 | 16.0        | 1036        | 609         | 24               |
| 13.00 | 18.3        | 1059        | 609         | 19               |



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**07Sep2023**

**Belt Data, DB Ver. = 2.18.0.0, Belt Rev. = 7/14/2016**

|                           |               |
|---------------------------|---------------|
| Belt Code                 | MT610         |
| # of Ribs / Cord Material | 8 / polyester |

### Tensioner Data

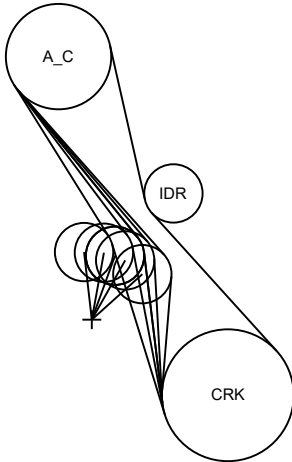
|                                                |               |
|------------------------------------------------|---------------|
| Type                                           | Automatic     |
| Design Tension N                               | 609 (137 lbf) |
| Inertia of the Tensioner Arm Kg-m <sup>2</sup> | 0.0076        |
| Tensioner Pulley Mass kg                       | 0.50          |

### Engine Data

|                                                |        |
|------------------------------------------------|--------|
| # of Cylinders                                 | 6      |
| Crankshaft Moment of Inertia kg m <sup>2</sup> | 0.1500 |

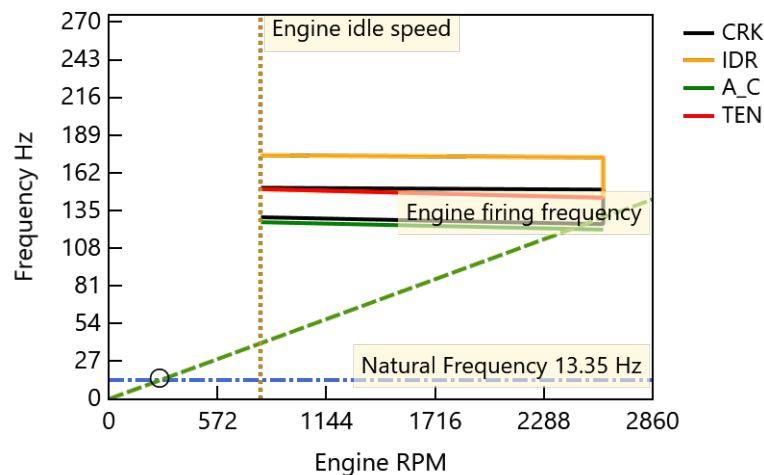
### Accessory Data

|     | File Name - if used | Moment of Inertia kg m <sup>2</sup> |
|-----|---------------------|-------------------------------------|
| IDR | IDR.IDR             | 0.0004                              |
| A_C | AC.A_C              | 0.0030                              |
| TEN | TEN.TEN             | 0.0076                              |



## RESULTS

### Natural Frequency Map



Note: Only low frequency spans are shown.  
System Resonance (Mode 1: 12.61)

Note: If Engine Idle is near system resonant RPM,  
then accessory drive system may have  
excessive vibration.  
If system resonance is within a span  
frequency window, then span vibration  
may have excessive amplitude.



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Drive Notes

| Date | Note |
|------|------|
|------|------|